

Thoracic endovascular aneurysm repair of a ruptured zone 0 mycotic thoracic aortic aneurysm

Valentyna Kostiuk^a, Prashanth Vallabhajosyula^b, Naiem Nassiri^c, Britt H. Tonnessen^{d,*}

^a Yale University School of Medicine, New Haven, CT, USA

^b Division of Cardiac Surgery, Department of Surgery, Yale University School of Medicine, New Haven, CT, USA

^c The Vascular Care Group, Darien, CT, USA

^d Division of Vascular Surgery and Endovascular Therapy, Department of Surgery, Yale University School of Medicine, 330 Cedar Str., Boardman 204, New Haven, CT 06510, USA

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ABSTRACT

A 48-year-old woman presented with an acute type B aortic dissection. Despite medical management with anti-impulse therapy in intensive care, she continued to have chest and back pain and underwent aortic arch debranching and thoracic endovascular aneurysm repair (TEVAR). Her post-operative course was complicated by sternal and mediastinal infection. Several weeks later, the patient developed acute onset chest pain and computed tomography angiogram revealed contrast extravasation just proximal to the endograft. Emergent zone 0 TEVAR was performed with exclusion of hemorrhage and continued filling of the debranched vessels. Post-operatively, fevers continued despite 6 weeks of antibiotic treatment and phage therapy was prescribed for 5 days. At 2-year follow-up, there was no recurrence of infection.

Introduction

Mycotic thoracic aortic aneurysms (MTAAs) are rare with prevalence of 0.6–2.6 % and pose high risk of expansion and rupture.^{1,2} Both open and endovascular repair are associated with high mortality and morbidity. The peri-operative mortality for MTAA repairs involving the aortic arch is about 25 % for TEVAR and 10 %–60 % for open repair.³ Open repair utilizes circulatory arrest or cardiopulmonary bypass and surgical resection of the infected aneurysm in conjunction with in-situ or extra-anatomic reconstruction.⁴ TEVAR has lower morbidity, but involves placement of a prosthetic graft into an infected aneurysm.⁵ Nonetheless, the re-infection rates after TEVAR for MTAA have been relatively low with freedom from re-infection reported as 81.2 % at 2 years.⁶ We present a TEVAR for a ruptured zone 0 MTAA complicated by multidrug-resistant *Klebsiella pneumoniae* infection requiring phage therapy.

Case description

The patient is a 48-year-old woman with medical history of coronary artery disease, congestive heart failure, diabetes, and hypertension. In 2015, she underwent repair of a symptomatic focal infrarenal aortic

dissection using a 28 × 82 mm Endurant II Iliac Extension stent graft (Medtronic, Minneapolis, MN). Then in 2020, she underwent an EVAR for symptomatic AAA using bifurcated modular Excluder® device (W. L. Gore & Associates, Newark, DE). Subsequently, she developed a type IB endoleak 7 months later due to enlargement of the iliac artery and underwent right iliac branched device repair with Gore Excluder® Iliac Branch Endoprosthesis. In 2021, she presented with an acute type B thoracic aortic dissection and recalcitrant chest pain despite medical management. She underwent a debranching of the aortic arch vessels and a TEVAR using RelayPro thoracic stent graft (Terumo Medical Corporation, Somerset, NJ). Her post-operative course was complicated by a sternal wound infection with multidrug-resistant *Klebsiella pneumoniae*, which was managed by sternal wound debridement, multiple chest washout procedures and an omental flap. Other post-operative complications included ventilator-associated pneumonia and bacteremia. Two weeks later, she developed fevers and sudden onset chest pain, and CTA demonstrated aortic arch contrast extravasation just proximal to the endograft (Fig. 1). Cardiac surgery deemed open repair to be prohibitive risk given prior operations, therefore endovascular repair approach was chosen for this patient. The debranching grafts were identified by previously placed radio-opaque markers, and a 3-cm landing zone proximal to the prior endograft in aortic zone 0 was

* Corresponding author.

E-mail address: britt.tonnessen@yale.edu (B.H. Tonnessen).

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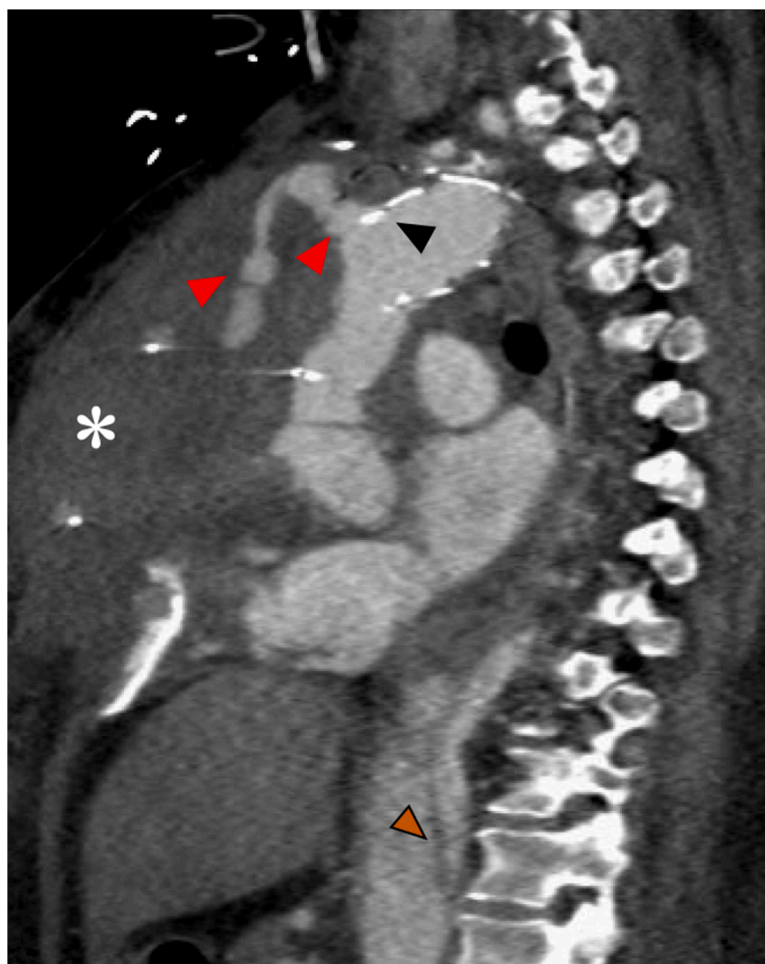


Fig. 1. CTA demonstrates a large stream of active extravasation (red arrowheads) just proximal to prior endograft placement (black arrowhead) with a significant mediastinal hematoma (*). Distal aortic dissection flap is marked (orange arrowhead).

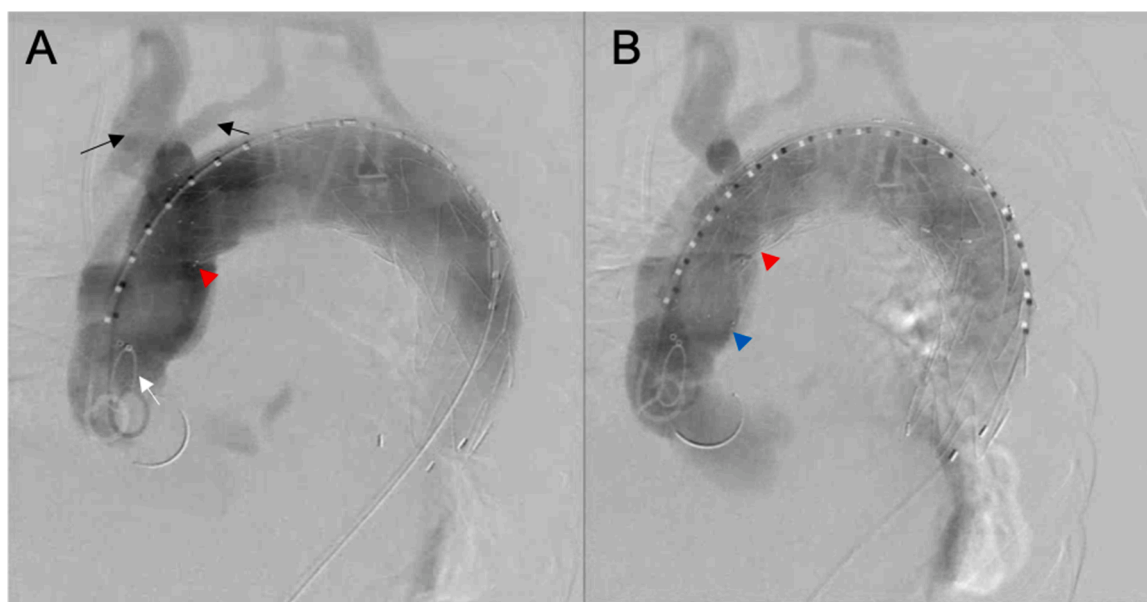


Fig. 2. (A) Intraoperative aortogram reveals ruptured proximal thoracic aorta in zone 0 with previous endograft (red arrowhead), arch debranching procedure (white arrow marks origin) with a 10 mm graft to the innominate artery and 8 mm graft to the left common carotid artery (black arrows) and superimposed area of hemorrhage. (B) Endograft was extended into zone 0 (blue arrowhead) with continued filling of innominate artery, left carotid artery and carotid-subclavian bypass vessels.

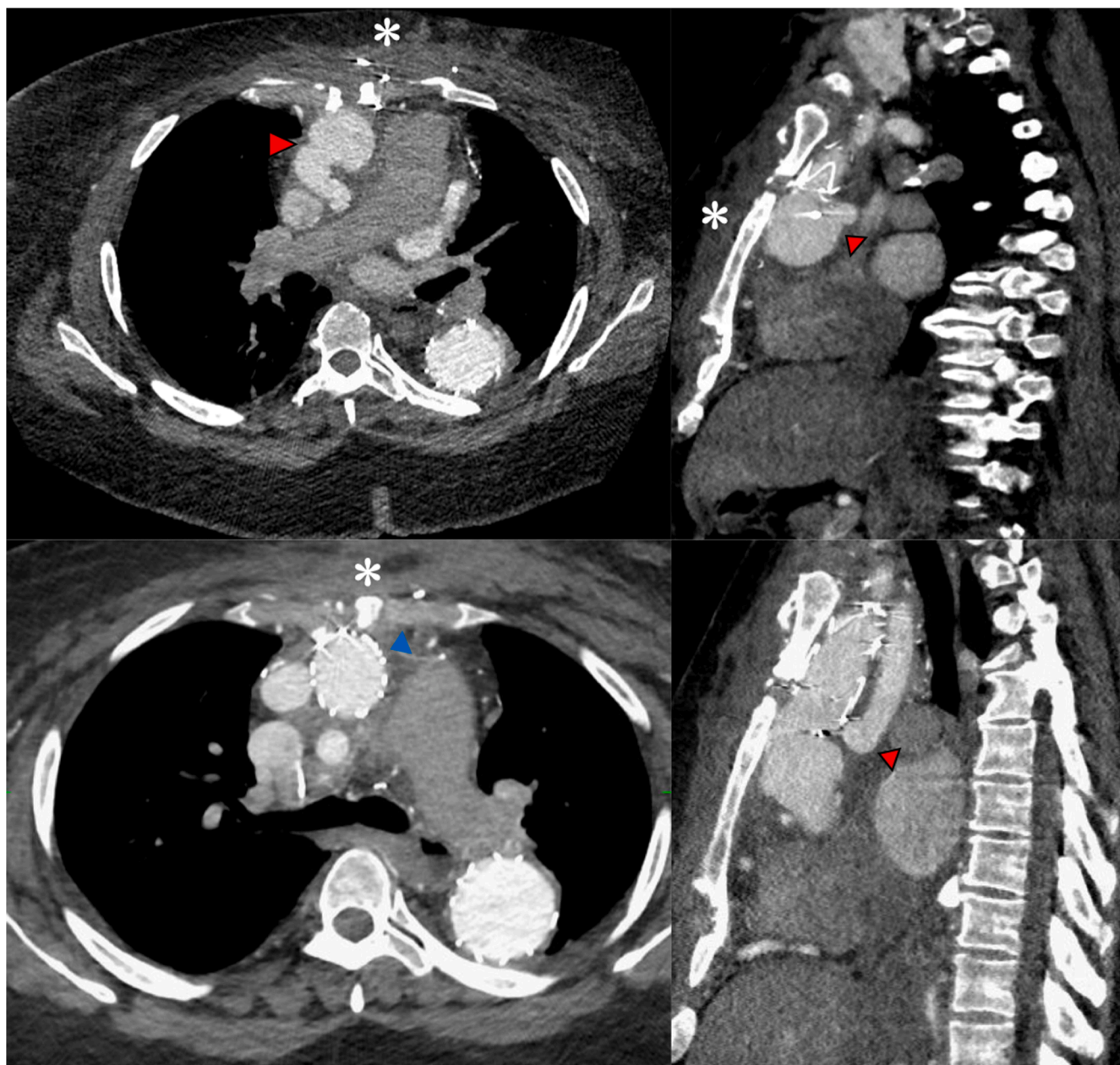


Fig. 3. CTA obtained 4 months post-operatively with axial and comparable level sagittal images demonstrate patent debranched vessels (red arrowheads), the absence of endoleak (blue arrowhead), and resolution of mediastinal hematoma (*).

utilized (Fig. 2A). TEVAR achieved exclusion of hemorrhage and continued filling of the debranched arch vessels (Fig. 2B). Post-operatively, patient continued to have fevers despite 6 weeks of antibiotic treatment and underwent FDA-approved phage therapy. Five daily treatments with the combination of intravenous (1×10^9 phage was diluted into 100 mL of sterile normal saline) and local therapy administered via drain or topically (1×10^9 phage cocktail diluted into 30 mL normal saline). Vital signs were monitored during treatments. Patient tolerated all treatments well without adverse reactions. In addition to phage therapy, patient was treated with intravenous ertapenem 1000 mg daily for 6 months. Following ertapenem therapy, patient was transitioned to oral omadacycline 300 mg daily but had to stop antibiotic treatment after 6 months due to insurance coverage loss. There was no endoleak on follow-up CTA (Fig. 3). Due to renal insufficiency, the majority of long-term imaging studies were done without contrast and demonstrated no evidence of recurrent infection at 2-years follow-up.

Discussion

The management of MTAA is challenging, and ruptured MTAA even more so. Mortality rates for thoracic aortic graft infections range between 25 % and 75 % (7) with at least 20 % risk of re-infection with open surgical repair.⁸ The complete resection of the infected aorta and graft, debridement of the surrounding infected tissue, graft replacement, and antibiotic regimen.⁹ TEVAR is an alternative treatment for patients who cannot tolerate open repair.⁷ Current literature consists of case reports and small series (Table 1). Frankel and colleagues reviewed their 30-year experience in a study of 75 mycotic aortic aneurysms.¹⁰ The majority of patients (81.3 %) underwent open repair, while hybrid and endovascular repairs were offered to patients who were not candidates for open surgery. Ascending thoracic aortic aneurysms were identified in 13 patients, of which 3 presented with rupture and required urgent surgery. There was a trend towards lower peri-operative mortality in the endovascular group when compared to open repair (23% vs 7 %, $p =$

Table 1
The summary of case series describing mycotic thoracic aneurysm repair and survival outcomes.

Author	Year	Number of Cases	Ascending/Descending Mycotic Aneurysm	Ruptures	Survival	Repair Approach
Frankel et al	2024	75	ascending thoracic aorta, aortic arch, descending thoracic aorta, thoracoabdominal aorta, and infrarenal abdominal aorta	26 (31 %)	55.7 % ± 5.8 % at 2 years; 39.0 % ± 5.7 % at 6 years; 26.9 % ± 5.5 % at 10 years	61 open repairs and 14 hybrid or endovascular repairs
Aoki et al	2017	26	thoracic aorta, thoracoabdominal aorta, and abdominal aorta	4 (15 %)	84.6 % overall	19 open repairs and 7 endovascular repairs
Sorelius, et al	2019	52	aortic arch and descending aorta	13 (25 %)	92 % at 30 days; 88 % at 3 months; 78 % at one year; 71 % at 5 years	50 endovascular and hybrid repairs and 2 open repairs
Patel et al	2010	27	thoracic aorta	None	57.8 % at 3 years	27 endovascular repairs
Kan et al.	2007	48	thoracic and abdominal aorta	18 (38 %)	89.6 % ± 4.4 % at 30 days; 82.2 % ± 5.8 % at 2 years	48 endovascular or hybrid repairs

0.2). Moreover, the authors reported high mortality of 38 % in all patients who presented with ruptures. Additionally, 17 % of patients developed aortic re-infection requiring reinterventions and overall survival at 2 years was 55.7 %.

A study from Hirosaki University in Japan examined 26 consecutive patients with infected thoracic, thoracoabdominal and abdominal aortic aneurysms between 2002 and 2015.¹¹ Of these, 19 patients were treated with open aneurysm repair, 6 patients underwent TEVAR, and 1 patient had an EVAR. In-hospital deaths occurred in 2 patients (7.7 %, 2/26) who underwent emergency total arch replacement for ruptured aortic arch aneurysms. Additionally, 2 patients who were initially treated with TEVAR required subsequent open revisions within 6 and 15 months to address a graft infection and a type I endoleak, respectively. In a Swedish study of 52 mycotic thoracic aneurysms treated between 2000 and 2016,¹² 6 aortic arch aneurysms were included, one of which subsequently underwent an open revision to address a graft infection and a concomitant type I endoleak. The safety and effectiveness of TEVAR for MTAA repair was examined in 27 patients treated between 1993 and 2007 at the University of Michigan.⁶ The results in this cohort demonstrated a technical success in 96.3 % of patients with in-hospital mortality of 11.5 %, overall mortality of 65 % (*n* = 14) with 4 deaths due to aortic re-infection, and freedom from re-infection of 81.2 % at 2-years.

The use of novel phage therapy for infections with antimicrobial resistance has been described.¹³ The regulatory framework for this therapy is currently provided by the Food and Drug Administration's Emergency Investigational New Drug guidelines.¹⁴ The strategy behind phage therapy involves using a virus to invade specific bacteria causing the disruption of bacterial metabolism and subsequent lysis.¹⁵ To produce phage preparation, the virus (phage) infects the bacteria, amplifies within their bacterial hosts, and causes bacterial lysis. This process releases high-titer phage solution that is purified from bacterial cell components through the process of centrifugation prior to injection into patient's bloodstream.¹⁶ Thus far, the use of phage therapy has been reported in 53 manuscripts focusing on 70 cases, the majority of which described the multidrug-resistant infections of implanted devices (30 %) and pulmonary system (17 %).¹⁷ Phage therapy was described in one patient with Dacron aortic graft infection complicated by an aorto-cutaneous fistula due to *Pseudomonas aeruginosa*.¹⁸ This patient had an aortic arch replacement procedure for an aortic aneurysm, which was complicated by *P. aeruginosa* mediastinal and aortic graft infection. Subsequently, the patient underwent multiple wound debridements and omental and bilateral pectoralis major flaps. However, the patient developed a mediastinal fistula and recurrent *P. aeruginosa* infection and was successfully treated with phage therapy and ceftazidime infusion.

In our case, aortic arch infection with drug resistant Klebsiella and

rupture was treated with emergent zone 0 TEVAR and subsequent phage therapy. At 2 years, the patient remained free of clinical and radiographic signs of re-infection.

Conclusion

MTAAs have high mortality and morbidity. Standard open repair is via sternotomy or thoracotomy with circulatory arrest. TEVAR is a consideration in high-risk patients for initial stabilization in combination with long-term with suppressive antibiotics. Additionally, the use of phage therapy to treat antibiotic-resistant infections in patients with MTAAs is intriguing, but currently only available in clinical investigations and for compassionate use.

Patient consent statement

Informed consent was obtained from the patient for publication.

Patient consent statement

The patient provided written consent for the publication of this case report.

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CRediT authorship contribution statement

Valentyna Kostiuk: Writing – review & editing, Writing – original draft, Project administration, Data curation, Conceptualization. **Prashanth Vallabhajosyula:** Writing – review & editing, Writing – original draft, Conceptualization. **Naiem Nassiri:** Writing – review & editing, Writing – original draft, Conceptualization. **Britt H. Tonnessen:** Writing – review & editing, Writing – original draft, Project administration, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Prashanth Vallabhajosyula reports a relationship with Terumo Aortic, Inc. that includes: consulting or advisory. Naiem Nassiri reports a relationship with Terumo Aortic, Inc. that includes: consulting or

advisory and speaking and lecture fees. Naiem Nassiri reports a relationship with Medtronic Aortic, Inc. that includes: consulting or advisory and speaking and lecture fees. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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